



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Storage

Objectives of this sheet

Initial contact with the Azure Storage service

Bibliography and useful links

Azure Storage documentation

<https://learn.microsoft.com/en-us/azure/storage/common/storage-introduction>



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Storage

In this assignment you will use your ipcbcampus student account to access Azure

At the end of the job, you must delete all Azure resources so that you can preserve your balance

1) Storage Account

Before creating storage services, you'll need to create the base, which is the Storage Account.

a) Create a new Storage Account with the following characteristics:

i) Basics

- Resource Group: class
- Storage account name: "**your_name**"**random number**", e.g. vitor783457
- Region: Central France
- Performance: Standard
- Redundancy: Zone-redundant storage

ii) Advanced

- Activate the option to allow "anonymous access on individual containers"
- Don't change anything in the other options

iii) Networking

- Don't change anything

iv) Data protection

- Activate all options except "Enable version-level immutability support"
- In the number of days settings, enter 1, 2, 2, 2 days respectively

v) Encryption

- Don't change anything

vi) Tags

- Don't change anything

b) Create the Storage Account and wait for it to be created.



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Storage

2) Blob Storage - key features

- a) In the Storage Account you've just created, create two containers for blobs (not to be confused with Docker Containers - these are just storage containers), one called "private-blobs" for private access and another called "public-blobs" for public access, with public access of type "Blob".

Home > vitor84576

vitor84576 | Containers ⚙️ ☆ ...

Storage account

Search << **+ Container** 🔒 Change ac

Search containers by prefix

Name

☐	\$blobchangefeed
☐	\$logs

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Data storage

- Containers**
- File shares
- Queues
- Tables

- b) Through the Azure portal, upload some files (these could be the slides from the lessons) to both blob containers.
- c) Choose one of the blobs (files) in the "public blobs" container and get its URL.



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Storage

Seguranca em Cloud Computing - azure.pdf ...

Blob

Save Discard Download Refresh Delete Change tier Acquire lease Break lease Give feedback

Overview Versions Snapshots Edit Generate SAS

Properties

URL <https://vitor84576.blob.core.windows.net/seguranca/Seguranca em Cloud Computing - azure.pdf>

LAST MODIFIED 5/4/2023, 10:03:43 AM

CREATION TIME 5/4/2023, 10:03:43 AM

VERSION ID 2023-05-04T09:03:43.2408144Z

TYPE Block blob

SIZE 3.45 MiB

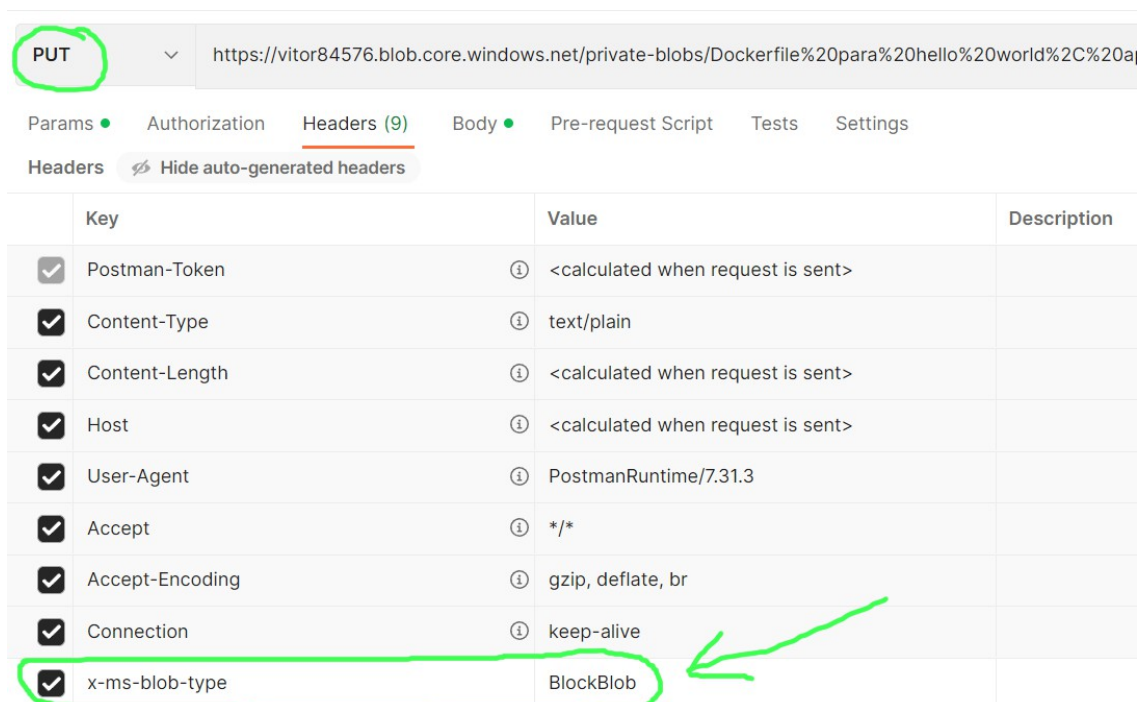
ACCESS TIER Hot (Inferred)

- Then, in another browser, try opening that URL. What happened?
- Now try the same thing, but with one of the blobs in the "private-blobs" container. What happened? Why did it happen?
- Now, generate a SAS - Shared Access Signature for the same blob, with the following characteristics:
 - Signing method: Account Key
 - Signing key: Key 1
 - Stored access policy: None
 - Permissions: Read, Write, Delete
 - Start Date: today, in 5 minutes
 - End Date: today, 10 minutes from now
 - Allowed protocols: HTTPS only
- Now, immediately use the Blob SAS URL (in another browser) and try to access the blob. What happened?
- Wait 5 minutes and try again. What happened?
- Wait 10 minutes and try again. What happened?
- How can this SAS functionality be useful?

Practical work on Storage

3) Blob Storage - access via REST API and version control

- Create a small text file text.txt on your computer and upload it to the private-blobs" container.
- Create a SAS that allows you to **read**, **create**, **write** and **delete** this blob
- Using the Postman application and the SAS created, use the HTTP PUT method to change the content of the blob, with a text of your choice (the text must be placed in the body of the HTTP request). You may have to add the following HTTP header:



- Send several different texts in at least 5 consecutive requests.
- In the Azure portal, check what the content of the blob is.
- Check if there are several versions of Blob. How many versions are there?
- Suppose you wanted to revert the blob to a specific version. How do you do that? Try it.



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

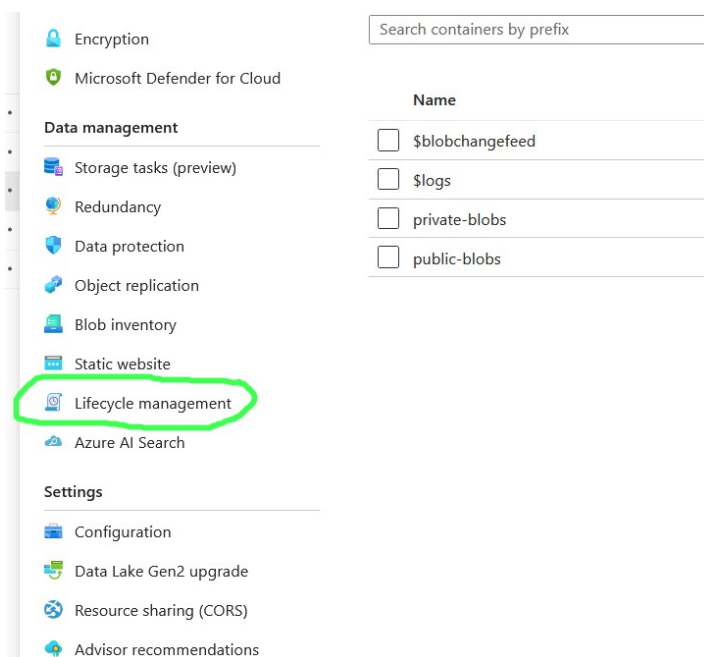
School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Storage

- h) Now use Postman to delete the blob - use the HTTP DELETE method. Did you manage to delete the blob using Postman?
- i) Suppose you want to recover the deleted blob. How do you do that? Try recovering it.
- j) Now take a snapshot of one of the other blobs. What's the difference between a snapshot and a version?

4) Tier management

- a) Select one of the blobs and move it from the "Hot" Tier to the "Cool" Tier. What does this mean in terms of **cost** and **speed of access**?
- b) The storage system makes it possible to automate the life cycle of blobs. See image:



- c) In lifecycle management, create rules that automatically:
 - i) Move the **base blobs** to Tier "cool" if they have not been modified in the last 30 days.
 - ii) Move the **snapshots** to Tier "cool" 10 days after they have been created.
 - iii) Move the **versions** to the "cool" Tier 5 days after they have been created and delete them after 30 days.



Instituto Politécnico
de Castelo Branco



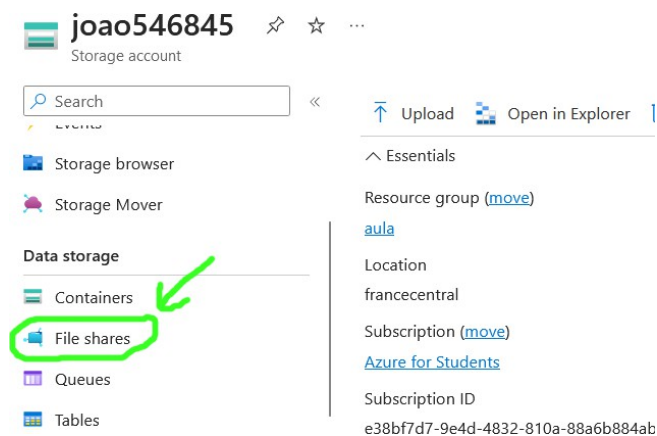
Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

School of Technology
Computer Engineering course
Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Storage

5) File Shares

a) In the Storage Account, create a **File Share** called "website", in Tier Hot.



b) Now upload an **index.html** file, with a message of your choice, to the File Share you created.

c) Create and run a Docker Container, with an NGINX web server, that mounts this File Share in the Document Root folder of the NGINX server, in order to use the File Share as the Document Root of the web server.

Tip: <https://learn.microsoft.com/en-us/azure/container-instances/container-instances-volume-azure-files>

d) Try viewing the website in a browser. Is your index.html page shown?

e) If you want to change the page, do you have to change or even stop the Docker Container? Try changing the contents of the index.html file in the file share.